

We denote by $\pi : G \rightarrow G/H$ the quotient map and let $\pi^* : L(G/H) \rightarrow L(G)$ be the map defined by $\pi^*(f) = f \circ \pi$.

Lemma 2.3 ([29], Lemma 4.5). *We have $\pi^* \left(\widehat{G/H} \right) \subseteq H^\perp$ and the restriction map $\pi^* : \widehat{G/H} \rightarrow H^\perp$ is an isomorphism of groups.*

2.2. Weighted graphs. The primary goal of this subsection is to introduce weighted graphs and record an inequality involving their eigenvalues. Standard references are [28, 27, 22]. Let $\mathcal{G} = (V, E, w)$ be an undirected, weighted and connected graph. Here V is the set of vertices, E is the set of edges and $\omega : V \times V \rightarrow \mathbb{R}_{\geq 0}$ satisfies $\omega(x, y) = \omega(y, x)$ for any $(x, y) \in V \times V$, and $\omega(x, y) > 0$ if and only if $\{x, y\} \in E$. The function ω is called the *weight function* of $\mathcal{G}$. Given $x, y \in V$, we say that x and y are *adjacent*, denoted as $x \sim y$, if $\{x, y\} \in E$. The *degree of a vertex* $x \in V$ is $d_x := \sum_{y \in V, y \sim x} \omega(x, y)$. We denote $d_G := \max_{x \in V} d_x$.

Let $L(V, \mathbb{R})$ be the vector space consisting of all real-valued functions on V . The *Laplacian operator* of $\mathcal{G}$ is the linear map $\Delta : L(V, \mathbb{R}) \rightarrow L(V, \mathbb{R})$ defined by

$$(2) \quad (\Delta u)(x) = \sum_{\substack{y \in V \\ y \sim x}} \omega(x, y)(u(y) - u(x)), \quad u \in L(V, \mathbb{R}), \quad x \in G.$$

We remark that $\Delta = A - D$ where $A := (\omega(x, y))_{x, y \in V} \in \mathbb{R}^{|V| \times |V|}$ and $D := \text{diag}(d_x)_{x \in V} \in \mathbb{R}^{|V| \times |V|}$.

Moreover, we define two bilinear operators on $L(V, \mathbb{R})$:

$$(3) \quad \Gamma(u, v) = \frac{1}{2} (\Delta(uv) - u\Delta v - (\Delta u)v), \quad \Gamma_2(u, v) = \frac{1}{2} (\Delta\Gamma(u, v) - \Gamma(u, \Delta v) - \Gamma(\Delta u, v)),$$

where $u, v \in L(V, \mathbb{R})$. For simplicity, we denote $\Gamma(u) := \Gamma(u, u)$ and $\Gamma_2(u) := \Gamma_2(u, u)$. We say that $\mathcal{G}$ is $\text{CD}(0, \infty)$ if $\Gamma_2(u) \geq 0$ for all $u \in L(G, \mathbb{R})$.

Theorem 2.4 ([28], Theorem 1.2). *Let $\mathcal{G} = (V, E, \omega)$ be an undirected, weighted and connected graph. Suppose that $\mathcal{G}$ is $\text{CD}(0, \infty)$ and $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_n$ are eigenvalues of $-\Delta$. Then for any natural number $k \geq 2$, we have $\lambda_k \leq C d_G k^2 \lambda_2$ where $C := \left(\frac{20\sqrt{2}e}{e-1} \right)^2$.*

3. POSITIVE DEFINITE FUNCTIONS

In this section, we investigate real-valued functions whose Fourier coefficients are all non-negative. These functions, known as positive definite functions, play a crucial role in harmonic analysis on groups [7, 8, 32]. To begin with, we summarize several basic properties of functions on finite abelian groups in the following lemma.

Lemma 3.1. *Given a function $f : G \rightarrow \mathbb{C}$, we denote by H the subgroup of $\widehat{G}$ generated by $\text{supp}(\widehat{f}) := \{\chi \in \widehat{G} : \widehat{f}(\chi) \neq 0\}$. Then the followings hold:*

- (a) *There is a well-defined function $\widetilde{f} : G/H^\perp \rightarrow \mathbb{C}$ such that $\widetilde{f}(xH^\perp) = f(x)$ for any $x \in G$.*
- (b) *Let $\pi : G \rightarrow G/H^\perp$ be the quotient map and let $\pi^* : \widehat{G/H^\perp} \rightarrow \widehat{G}$ be the induced map defined by $\pi^*(\rho) := \rho \circ \pi$ is an isomorphism from $\widehat{G/H^\perp}$ to H .*
- (c) *$\widetilde{f} = \sum_{\chi \in \text{supp} \widehat{f}} \widehat{f}(\chi)(\pi^*)^{-1}(\chi)$.*
- (d) *Suppose that f is real-valued. For any $\chi \in \widehat{G}$, we have $\widehat{f}(\chi^{-1}) = \overline{\widehat{f}(\chi)}$. If moreover $\widehat{f}(\chi) \in \mathbb{R}$, then $\widehat{f}(\chi^{-1}) = \widehat{f}(\chi)$.*

Proof. To prove (a), it is sufficient to show $f(y) = f(x)$ for all $y \in xH^\perp$. By definition, there exists some $z \in H^\perp$ such that $y = xz$. Since H is generated by $\text{supp}(\widehat{f})$, $H^\perp = \{z \in G : \chi(z) = 1, \chi \in \text{supp}(\widehat{f})\}$. This implies

$$f(y) = \sum_{\chi \in \text{supp}(\widehat{f})} \widehat{f}(\chi)\chi(y) = \sum_{\chi \in \text{supp}(\widehat{f})} \widehat{f}(\chi)\chi(x)\chi(z) = \sum_{\chi \in \text{supp}(\widehat{f})} \widehat{f}(\chi)\chi(x) = f(x).$$